

PART TWO

SOME GENERAL OBSERVATIONS

By this point we have studied a number of the essential features of the elements of complex function theory and have become acquainted with the simplest basic concepts and tools of the Cauchy theory used daily by workers in other fields of mathematics for whom complex function theory is an important auxiliary discipline. We have by no means exhausted the study of the elements. In particular, we have not touched analytic continuation nor the notion of a Riemann surface. For a proper understanding of global complex function theory and its applications these notions are essential (they are indispensable in the study of differential equations in the complex domain). They will be treated in this part of the book.

In addition, this part will introduce topics that round out and give further applications of the theory developed in Part One without, as a rule, demanding the serious use of ideas not current at this phase of the reader's mathematical development. However, certain topics (such as vector-valued analytic functions) will require a small amount of information of a non function-theoretic nature and the appropriate background will be sketched.

The material treated in Part Two will range from simple remarks and problems to the exposition of basic topics. The emphasis placed in Part One on independent work and problems will persist.